# Sofiat Adeyemi

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**Portfolio - sofiaade.github.io**

## EDUCATION

**Southeastern Louisiana University (SLU)** Hammond, LA
Bachelor of Science in Computer Science, Concentration: Data Science December 2026

- Thesis: Fine-Tuning LLMs using Reinforcement Learning

## RELEVANT COURSEWORK

**Core CS:** Algorithms Design and Implementation, Software Engineering, Computer Architecture, Calculus, Linear Algebra, Statistics, Computing Systems
**Data Science:** Machine Learning, Database System, Data Mining, Artificial Intelligence, Computer Vision

## RESEARCH EXPERIENCE

**GPT-2 Model Training from Scratch** – ***Stage 1 of Thesis - Deep Learning Research Project***

• Trained and optimized GPT-2-style language models from scratch using PyTorch and Hugging Face Transformers.
• Implemented tokenization, attention mechanisms, and gradient checkpointing for efficient model scaling.
• Followed deep learning best practices from Sebastian Raschka's *Deep Learning with PyTorch* series to improve convergence and model interpretability. We also used Sebastian Raschka's LLM from scratch on github.
• Conducted hyperparameter tuning, evaluation, and deployment using MLflow and AWS.

## PROJECT EXPERIENCE

**BookManagement - Online BookStore Database** Hammond, LA

- Using SQL, we developed a database model for an online bookstore that allows users to browse, purchase, and review books.
- The database system will store book details and track purchase history. Book inventory will automatically update after sales, including when new stock is received.

**Mood Classification with Machine Learning** Hammond, LA

- Using Python and related libraries like Scikit Learn, we developed machine-learning models using ANN, SVM, DT, and KNN to predict the class of object labels.
- The process involved data preprocessing, feature extraction, model training, and evaluation to determine the most effective classification approach

**TaskMaster – Time Management Web App** Hammond, LA

- Served as a project team leader and developed a time management website.
- Incorporated C #, # ASP.NET, React, Typescript, Web API, Entity Framework, and semantic ui to create an interactive website

**Student Wellbeing Data Analysis** Hammond, LA

- Served as a project team leader, and we did an in-depth analysis of student issues.
- Incorporated Python libraries (NumPy, pandas, matplotlib, seaborn) to identify key issues students face in different districts and to understand student groups based on ethnic and social conditions.

**Fraud Detection using Unsupervised Learning**

- Built unsupervised **K-Means** and **Hierarchical Clustering** models to detect anomalous financial transactions.
- Cleaned and visualized data using **pandas**, **NumPy**, **matplotlib**, and **seaborn**.
- Identified unusual transaction patterns indicating potential fraud.

**Traffic Sign Recognition Using A CNN made from scratch** Hammond, LA

- Build a CNN architecture from scratch without using pre-trained models
- Achieve a classification accuracy above 93% on the test dataset
- Address real-world challenges like varying lighting conditions, weather effects, and partial obstructions
- Develop a strong model that can generalize to new traffic sign images

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## TECHNICAL SKILLS

- **Programming Languages**: SQL, C, C#, Java, Python, TypeScript
- **Deep Learning:** PyTorch, TensorFlow, Keras, Hugging Face Transformers,
- **Machine Learning:** Scikit-learn, MLflow, OpenCV, Gradient Boosting, Feature Engineering
- **Data Science & Visualization:** Pandas, NumPy, MNE, Seaborn, Matplotlib
- **Web/Systems:** React, FastAPI, Flask, ASP.NET Core, REST APIs, Git, Multithreading, IoT, Docker

- **Cloud & Tools:** AWS, Google Colab, Kaggle, Jupyter Notebook, VS Code, Entity Framework.
- **Soft Skills:** Leadership, Communication, Technical Writing, Team Collaboration, Problem Solving, Adaptability

## WORK EXPERIENCE

**Computer Science Tutor** Hammond, LA
*SLU Computer Science Department* August 2025 – Present

## HONORS & INVOLVEMENT

- **Dean's List** - Fall 2023, **Honors List** - Spring 2024, **President's List** - Fall 2024, Spring 2025
- **ACM Member** - Fall 2023 - present